

Right-Sized AI at the Edge: Reproducible Measurements and Conditional Sustainability Analysis

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Abstract—Moving language-model inference to consumer hardware can change its energy, carbon, water, and financial costs, but a smaller per-request number does not establish a saving at equivalent service quality. We examine archived single-stream measurements from a Radeon 8060S system and an RTX 5070 system and provide a boundary-explicit accounting model. An independent implementation replays 68 stored generation repetitions, including 38 requests with bracketed board-power telemetry. At 499 input and 300 output tokens, the RTX 5070 records median request-window board energies of 0.091 Wh and 0.19 Wh for the recorded Ministral-3 3B and 8B builds. A second session on the same machine gives the same rounded medians. Assuming 60 W of additional host power and 90% supply efficiency changes the primary-session estimates to 0.13 Wh and 0.27 Wh; these are scenarios, not wall-meter observations. A twelve-row throughput/file-size comparison demonstrates why this proxy cannot be read as measured memory-bandwidth utilization. We derive accounting expressions that include failed local attempts, cloud fallback, and a common demand-growth factor across footprint metrics. The records do not establish quality-matched cloud displacement, population-wide task coverage, total cost of ownership, or fleet-scale environmental savings. The contribution is a reproducible deployment case study and a specification of the additional evidence needed for a credible edge-versus-cloud sustainability claim.

Index Terms—edge inference, energy measurement, carbon accounting, water consumption, small language models, reproducibility

I. INTRODUCTION

Whether an inference deployment is sustainable depends on the service it delivers and the boundary used to count its resources. Board energy, whole-computer electricity, data-centre electricity, and life-cycle emissions are different quantities. A short local response is not an interchangeable substitute for a long cloud response without a task-level quality test.

The International Energy Agency’s 2025 base case projects approximately 945 TWh of *all data-centre* electricity demand in 2030 [1]. That motivates careful measurement; it does not supply a direct estimate of the fraction that consumer inference could replace. We therefore ask what can be established from single-stream consumer deployment records, and what remains conditional when they enter an environmental comparison.

The supplied artifact contains measurements recorded in September 2026 on two Windows 11 machines. We retain their device and model labels as recorded, without treating a runtime tag as an immutable model identifier or independent

verification of a vendor release. No new hardware experiment was performed for this revision.

Our contributions are:

- 1) an arithmetic replay of stored timings and power traces, distinguishing raw observations from aggregate-only records;
- 2) request-window energy results with consistent board, host, and supply boundaries, and a cautionary analysis of file-size proxies;
- 3) conditional accounting that charges local failures and uses one demand-growth assumption for energy, water, carbon, and cost.

We do not claim a new inference algorithm, the first consumer-GPU benchmark, or an experimentally demonstrated reduction in cloud impact.

II. RELATED WORK AND COMPARISON BOUNDARIES

Elsworth et al. measure production AI serving at Google and report 0.24 Wh for a median Gemini Apps text prompt, accounting for accelerator, host, idle-capacity, and facility energy [2]. Their production population is not the prompt used here. Jegham et al. combine API performance with inferred infrastructure in an environmental *estimation* framework [3]. Their numbers are not direct wall-power readings from those providers.

Consumer inference is already studied. Knoop and Holtmann evaluate consumer Blackwell deployments across models and configurations [4]. Tummalapalli et al. study sustained edge inference and explicitly discuss inconsistent power-measurement boundaries across platforms [5]. Our small case study complements these works through replayable request traces and boundary sensitivity; it does not supersede their broader evaluations. MLPerf Inference illustrates the importance of explicit workload, scenario, and quality requirements [6]. This is not an MLPerf submission.

Belcak et al. motivate heterogeneous small/large-model agents [7]. Their Appendix B gives *estimates* of replaceable calls in three agent architectures, not a sampled estimate of local success for our application. Luccioni et al. discuss rebound effects and the limits of inferring societal impact from technical efficiency [8]; that work does not calibrate a 15–25% take-back rate for this deployment. Water accounting must

TABLE I
ARCHIVED EVIDENCE. FILE NAMES SHARE THE SUFFIX `.JSON`. RAW
CONCERNS MAIN GENERATION REPETITIONS, NOT ALL AUXILIARY
OBSERVATIONS.

File stem	Models	Reps.	Output	Raw
amd-8060s-strixhalo	3	5	192	Yes
amd-8060s-strixhalo-ladder	9	7	128	No
rtx5070	2	5	192	Yes
rtx5070-study	2	7	300	Yes
rtx5070-replication	2	7	300	Yes
rtx5070-ladder	3	5	192	Yes

distinguish withdrawal from consumption, and on-site cooling from electricity-generation impacts [9].

III. DATA AND METHODS

A. Deployment Records and Protocol

The AMD records identify a Ryzen AI MAX+ 395 with Radeon 8060S graphics, nominal 64 GiB system memory, and Ollama 0.33.2. The NVIDIA records identify an RTX 5070, nominal 12 GiB graphics memory, an Intel Core i9-13900 host, Ollama 0.33.3, and driver 610.47. These are recorded configurations, not controlled comparisons of silicon architectures.

The harness uses a synthetic explanatory prompt, temperature zero, a fixed seed, a 4096-token context setting, and a discarded warm-up. Raw mode bypasses prompt templating [10]. A varying prefix is placed before the prompt to reduce prefix-cache reuse. These controls reduce a known confound; they do not prove that every runtime cache is disabled. Loading and first-use initialization are outside the warm request window.

Table I identifies the six files. The primary and repeat NVIDIA sessions retain seven repetitions per model, with 499 input tokens and 300 generated tokens. Other files have different prompt lengths and generation limits and are not pooled into those energy medians. The two Ministral-3 tags describe Q4_K_M builds, with recorded parameter labels of 3.8B and 8.9B. The short 3B and 8B labels below refer to their runtime names.

There are 68 stored main-run repetitions. The AMD ladder instead retains nine medians, ranges, and metadata records, each reporting seven repetitions. Those underlying repetitions are unavailable. Prefill sweeps also retain point medians rather than all individual timing observations. Two AMD ladder model labels are anonymized and their weights are not redistributed. Arithmetic checks and anonymized tags do not establish independent reproducibility of those builds.

B. Telemetry Integration

For NVIDIA, the harness polls `power.draw.instant` through `nvidia-smi`. NVIDIA describes this as the latest board-power measurement, not a whole-system meter [11]. Samples carry request-relative timestamps and bracket the request window.

For samples (t_i, P_i) in seconds and watts, define the linear interpolant

$$\hat{P}(t) = P_i + \frac{P_{i+1} - P_i}{t_{i+1} - t_i}(t - t_i). \quad (1)$$

For a request lasting T seconds, board energy is

$$E_b = \frac{1}{3600} \int_0^T \hat{P}(t) dt \quad \text{in Wh.} \quad (2)$$

The independent replay clips segments to $[0, T]$, rejects non-monotonic or unbracketed traces, and checks serialized energy, mean power, wall-window duration, token counts, and median throughput. It does not import the original analytical model.

Repeated polling does not create independent sensor readings. In the primary and repeat 300-token sessions, individual requests contain only 3–4 distinct in-window power values for 3B and 6–7 for 8B. These counts do not determine sensor accuracy or true update frequency. Instrument latency, smoothing, and interpolation can bias short-window estimates. We use two significant figures for energy as a conservative reporting convention, not a calibrated uncertainty bound.

C. Host and Supply Boundary

For assumed non-GPU host DC power P_h and supply efficiency η , a request-window AC estimate is

$$E_{AC} = \frac{E_b + P_h T / 3600}{\eta}. \quad (3)$$

We evaluate $P_h \in \{0, 30, 60, 100\}$ W with $\eta = 0.90$. Each run is transformed before taking the median; multiplying separate medians is not generally equivalent. Even the zero-host scenario includes supply loss and differs from the board-only value. These host and efficiency values are assumptions, not measurements. Display power, external networking, building cooling, and additional keep-awake time are outside this table.

An incremental-energy claim requires an idle counterfactual at the *same* measurement boundary. Subtracting whole-desktop idle watts from GPU-board watts is invalid. Already owning a device does not establish zero incremental idle time or zero life-cycle impact.

D. Timing Fits and File-Size Proxy

Generation throughput is $\tau = n_o / t_{\text{decode}}$, using server-reported output count and decode duration. For each AMD model, ordinary least squares fits four archived prompt-sweep medians:

$$\hat{t}_{\text{prefill}}(n_i) = a + bn_i, \quad \hat{\beta}(n_i, n_o) = \frac{a + bn_i}{n_o / \tau}. \quad (4)$$

$1/b$ describes a marginal fitted prefill rate, not end-to-end throughput. $\hat{\beta}(500, 300)$ is a model-based projection within the swept prompt range, not a measured 500/300-token AMD request.

The ladder statistic is

$$U_{\text{file}} = \frac{\tau W_{\text{file}}}{B_{\text{ref}}}. \quad (5)$$

TABLE II
RTX 5070 BOARD TELEMETRY FOR 499 INPUT AND 300 OUTPUT TOKENS.
SEVEN RUNS PER ROW. RANGES ARE OBSERVED MINIMA AND MAXIMA,
NOT CONFIDENCE INTERVALS.

Build	Session	Tok/s	Median Wh	Range Wh
3B	Primary	195.6	0.091	0.078–0.093
3B	Repeat	196.4	0.091	0.078–0.094
8B	Primary	101.8	0.19	0.18–0.19
8B	Repeat	102.5	0.19	0.17–0.20

TABLE III
PRIMARY-SESSION ENERGY IN WH. BOARD IS MEASURED VIA TELEMETRY;
ALL AC COLUMNS ARE EQ. (3) SCENARIOS WITH $\eta = 0.90$ AND THE
STATED HOST DC POWER.

Build	Board	AC: 0 W	AC: 30 W	AC: 60 W	AC: 100 W
3B	0.091	0.10	0.12	0.13	0.15
8B	0.19	0.22	0.24	0.27	0.31

W_{file} is recorded decimal GB, and B_{ref} is a nominal bandwidth reference, 256 GB/s for AMD and 672 GB/s for NVIDIA, as recorded in the artifact. We do not measure sustainable bandwidth. The numerator is *not* measured memory traffic: a file can contain inactive experts, vision components, metadata, and tensors accessed at different frequencies. Cache reuse, quantization, runtime kernels, and speculative decoding further affect the relationship.

IV. RESULTS

A. Arithmetic Replay and Request Energy

All 38 archived powered requests reproduce serialized energy within 5.0×10^{-7} Wh, consistent with rounding. This demonstrates internal arithmetic consistency, not physical calibration or authenticity. Likewise, $\bar{P}T = 3600E_b$ is an identity when $\bar{P}$ is derived from the same integral, not independent validation.

Table II shows stable rounded medians across two sessions. The file called replication is a repeat on the same machine and stack, not replication by an independent laboratory. The primary 3B and 8B window medians are approximately 1.62 and 3.11 s. Across primary and repeat sessions, wall time minus reported prefill and decode time ranges from about 26 to 72 ms. We do not attribute that residual to one mechanism.

At 60 W host power, modeled AC energy is 0.13 Wh for 3B and 0.27 Wh for 8B (Table III and Fig. 1). Neither can be compared directly to a production cloud median without matching the task and service boundary. The 192-token NVIDIA pilot remains in the replay but is not another repetition of the 300-token experiment. There is no AMD power trace; no AMD energy result is inferred from throughput alone.

B. AMD Throughput and Prefill

Decode medians are 78.4, 37.8, and 13.8 tokens/s (Table IV). Projected prefill/decode ratios are 0.050–0.078. These results describe one prompt family and serving configuration, not arbitrary agent workloads. The 27B tag has a different rate in

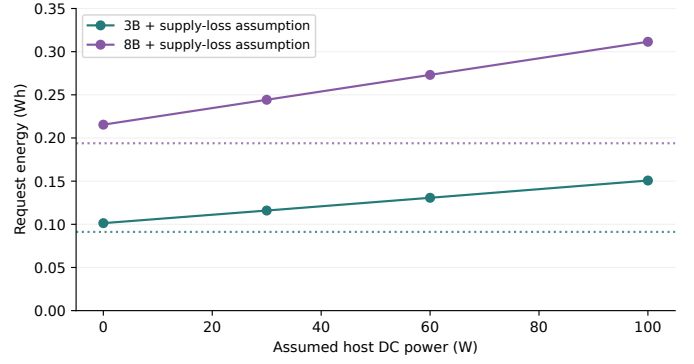


Fig. 1. Primary-session host-power sensitivity. Dotted lines are board medians. Solid curves include assumed host power and supply loss, not additional idle time or cooling.

TABLE IV
AMD MAIN-RUN DECODE AND RECONSTRUCTED PREFILL FITS. FITTED $1/b$
IS IN TOKENS/S AND a IN MS; $\hat{\beta}$ IS EVALUATED AT 500 INPUT AND 300
OUTPUT TOKENS.

Recorded build	Decode	$1/b$	a	$\hat{\beta}$
Ministral-3 3B	78.4	2711	5.7	0.050
Ministral-3 8B	37.8	1161	7.4	0.055
Qwen3.8 27B	13.8	341	233.2	0.078

the shorter ladder protocol; reporting both does not make them interchangeable.

C. Why the Ladder Is Not a Bandwidth Measurement

Table V retains all nine AMD and three NVIDIA rows. AMD proxies range from 76.7% to 581.6%; NVIDIA rows range from 63.1% to 93.2%. Values above 100% do not demonstrate violation of a physical limit. They reject the interpretation that file size equals externally transferred bytes per output token under the stated bandwidth reference.

An anonymized vision-language build reaches 105.7%. A dense runtime label does not imply every file byte is read from external memory per decoded token. Likewise, an architecture metadata flag for multi-token prediction does not prove its use or establish an acceptance rate. The excess supplies no upper bound on proxy error for other builds. Neither a 30-percentage-point separation between selected groups nor high proxy values establish causal architectural classification or universal memory-bound behavior. Hardware traffic counters and runtime instrumentation would be needed to test those explanations.

V. CONDITIONAL SUSTAINABILITY ACCOUNTING

A. Functional Unit and Quality

The desired functional unit is a completed task satisfying a predeclared quality and latency requirement, including retries and fallbacks. Token counts alone do not define that unit; different tokenizers and answers also matter. Our timing prompt is not a quality evaluation.

TABLE V

ALL ARCHIVED LADDER ROWS. PROXY PERCENTAGES ARE $100U_{\text{file}}$, RECOMPUTED FROM STORED FILE SIZES AND THROUGHPUT. SUMMARY MEANS THE RAW REPETITIONS ARE UNAVAILABLE. RUNTIME TAGS IDENTIFY THESE RECORDS, NOT UNIVERSALLY REPRODUCIBLE WEIGHTS.

Platform	Recorded model tag	File GB	Tokens/s	Proxy (%)	Evidence
AMD	private-3.1b-q8:latest	3.285	59.77	76.7	Summary
AMD	private-7.6b-vlm-q8:latest	9.453	28.62	105.7	Summary
AMD	gemma4:26b	17.988	58.34	409.9	Summary
AMD	gemma4:e4b	9.608	56.42	211.8	Summary
AMD	ministral-3:3b	2.954	78.62	90.7	Summary
AMD	ministral-3:8b	6.022	37.81	88.9	Summary
AMD	muse-glimmer:30b	18.157	12.46	88.4	Summary
AMD	qwen3.6:latest	23.938	62.20	581.6	Summary
AMD	qwen3.8:27b	17.742	19.63	136.0	Summary
RTX	ministral-3:3b	2.954	199.28	87.6	Raw
RTX	ministral-3:8b	6.022	103.96	93.2	Raw
RTX	qwen3:4b	2.497	169.92	63.1	Raw

For task classes k , let w_k be deployment frequencies, p_k the fraction attempted locally, and q_k the conditional probability that a local attempt is followed by a cloud call. Let $F_{e,k}$ and $F_{c,k}$ be per-attempt footprints for metric F , with comparable boundaries. If one cloud call resolves each escalation, expected footprint is

$$D_F = \sum_k w_k [(1 - p_k)F_{c,k} + p_k F_{e,k} + p_k q_k F_{c,k}] + R_F + N_F, \quad (6)$$

$$F_0 = \sum_k w_k F_{c,k}, \quad \sum_k w_k = 1. \quad (7)$$

R_F and N_F are routing and additional network overhead per original task, excluding anything already in the attempt footprints. Multiple retries require extra terms. The $p_k q_k F_{c,k}$ term must not be omitted merely because the final answer comes from the cloud.

In a one-class, no-failure special case, $D_F = fF_e + (1 - f)F_c + R_F + N_F$. We do not estimate f , p_k , or q_k from timing records. The earlier assumption $f = 0.82$ at relative-quality threshold 0.70 was not a measured routing rate; allowing 70% of a benchmark score is not evidence of equivalent service quality.

B. Carbon, Water, and Cost

With energy in Wh and location-based grid intensity I_j in gCO₂e/kWh, emissions are

$$C_j = E_j I_j / 1000 + A_j, \quad (8)$$

where A_j consistently allocates non-operational emissions in gCO₂e per task. Allocating fraction a of hardware embodied emissions M kgCO₂e over Q tasks gives $A = 1000aM/Q$. Allocation is not automatically consequential emissions avoided by using existing hardware. Purchase, lifetime, and replacement counterfactuals must be specified on both sides. Market-based provider values must not be silently mixed with location-based household factors.

If E_j includes facility electricity, PUE is known, and both water intensities describe consumption, operational water is

$$W_j = E_j (u_j / \text{PUE}_j + v_j) \quad \text{in mL.} \quad (9)$$

u_j is on-site L/kWh of IT electricity; v_j is off-site L/kWh of facility electricity. The factors of 1000 converting Wh to kWh and L to mL cancel. Withdrawal must not be added to a consumption-only input [9]. A zero on-site term is justified only if that boundary has no consumptive cooling. Electricity, building cooling, and manufacturing can still use water.

At electricity price p_{elec} in dollars/kWh,

$$K_{\text{electricity}} = E_{\text{AC}} p_{\text{elec}} / 1000. \quad (10)$$

The primary 8B, 60 W-host scenario at an assumed \$0.16/kWh gives approximately \$0.000044 per request for electricity. This excludes equipment, maintenance, standby allocation, network charges, and labor. It is not total cost of ownership or a quality-matched API saving.

C. Rebound and Break-Even Conditions

For a scenario with demand growth g , fixed task mix, and fixed per-task footprints, the *same* g applies to every metric:

$$D_F^{\text{eff}} = (1 + g)D_F, \quad S_F = 1 - (1 + g)D_F / F_0. \quad (11)$$

For $0 < D_F < F_0$, break-even growth is

$$g_F^* = F_0 / D_F - 1. \quad (12)$$

If $D_F \geq F_0$, there is no saving at nonnegative growth. Thresholds differ by metric because footprints differ, not because one scenario can have different query-volume growth for water and carbon.

A numerical unit check does not assert deployment parameters. Set $F_c = 1$, $F_e = 0.25$, $p = 0.80$, $q = 0$, and overhead zero. Then $D_F = 0.40$ and $g = 0.25$ gives $S_F = 0.50$; break-even is $g^* = 1.50$. If 10% of local attempts also invoke the cloud ($q = 0.10$), $D_F = 0.48$ and savings at the same growth fall to 0.40.

A take-back fraction r satisfies $g = r(F_0 - D_F) / D_F$ for a *single* metric with positive gross saving. Applying one r

separately to energy, water, and carbon generally implies different g values, not one physical demand scenario. No observed elasticity is available here; rebound is not empirically refuted.

D. Why We Do Not Report a Cloud-Savings Percentage

The 0.24 Wh Gemini median is a production-population measure with a broader boundary [2]. GPT-4o short and long estimates of 0.421 and 1.788 Wh are version-specific values from Jegham et al., v3, Table 4 [3]. Their workload definitions are 100/300 and 10 000/1500 input/output tokens. Neither supplies a quality-matched 499/300-token cloud counterfactual for these builds. Linear token-count normalization would ignore prefill, decode, caching, batching, and reasoning differences.

Google’s comprehensive breakdown includes host and infrastructure costs. Their existence does not mean a desktop removes them: it also has host and supply costs. Using a global grid factor does not reproduce a provider’s reported market-based carbon footprint.

Multiplying 945 TWh by assumed AI share, inference share, adoption, and a mismatched per-query reduction does not establish an IEA-backed fleet-saving forecast. It needs workload-weighted energy shares and displacement evidence. Monte Carlo over selected priors cannot supply that evidence; favorable simulation draws are not a measured probability that a deployment will save resources.

VI. TRANSPORT AND SYSTEM SCOPE

WebRTC data channels use SCTP over DTLS [12]. An ordinary TURN relay forwards encrypted traffic and need not receive application plaintext, provided endpoint authentication and signaling integrity hold. The security architecture distinguishes passive protection from an active attack through compromised or untrusted signaling [13]. An inference peer can read the prompt, as can a remote router that classifies it. Encryption does not remove metadata exposure or endpoint trust.

These are protocol considerations, not an implementation security audit. There is no measured network-energy experiment, relay probability study, or P2P-versus-cloud session trial here. We include network and routing as unmeasured terms in Eq. (6), without a numerical transmission-saving claim.

VII. LIMITATIONS AND REQUIRED EXPERIMENTS

The strongest supported claim is narrow: the traces are internally consistent and describe short, warm, single-stream runs on the stated stacks. Major limits remain.

External validity. Two machines, a synthetic prompt family, small repetition counts, and fixed output limits do not represent an assistant workload. Same-machine repeats do not establish cross-machine or independent reproducibility. Software, thermal state, clocks, model format, and batching can change results. Runtime tags can change; immutable digests and retrieval instructions are needed for full build reconstruction.

Measurement validity. Board telemetry is not wall energy. There is no external-meter calibration or systematic sensor-error

interval. Run ranges omit shared systematic errors. Auxiliary prompt sweeps and all AMD ladder rows lack their individual observations in this archive.

Service validity. There is no matched quality evaluation, routing confusion matrix, success/fallback trace, or representative task distribution. We cannot infer that parameter count alone determines capability, that all large local models lose to cloud serving, or that fine-tuning changes coverage by a known amount. The companion adaptation manuscript is outside this experiment.

The next decisive experiment would use a released, held-out task set with prespecified quality and latency thresholds. It would measure the same tasks on local and cloud alternatives, log failed attempts and retries, calibrate longer wall-meter runs against telemetry, and separate cold, active, and keep-awake energy. Multiple sessions and machines, randomized order, and thermal/clock records would improve uncertainty analysis. Cloud energy needs a cooperating provider’s instrumentation or explicit identification as an estimate; client API timing cannot measure provider power.

VIII. REPRODUCIBILITY AND RESEARCH INTEGRITY

The original harness and supplied records are in the public research repository [14]. The accompanying revision artifact adds `review/audit.py`, synthetic unit tests, a JSON report, and SHA-256 hashes of all six input files. Replay requires no live model, cloud credentials, private logs, or personal user data. It generates the numerical macros used by this manuscript and the host-sensitivity figure. From the research repository, run:

```
python review/audit.py
python -m unittest discover -s review
python -m unittest measure.test_bench
python review/build_paper.py
```

Figure generation requires `requirements.txt`. Legacy scenario generators and the companion manuscript are retained for provenance, but are not the validation path for these revised results.

An AI assistant assisted this revision with source checking, arithmetic replay, code, and drafting. This is not independent peer review or laboratory replication. Authors remain responsible for the measurements, interpretation, rights, authorship, and final submission declarations.

IX. CONCLUSION

The records support a reproducible, boundary-limited consumer-inference case study. For the 499/300-token RTX 5070 experiment, board telemetry yields medians of 0.091 and 0.19 Wh; host and supply assumptions materially increase system estimates. File-size throughput ratios cannot establish actual bandwidth utilization. A credible sustainability comparison must additionally measure equivalent task quality, failed local attempts, fallback, and comparable boundaries. Right-sizing remains a testable deployment hypothesis, not a quantified environmental benefit established by these records alone.

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